

Chapter 4

Solving for the Unknown

Math for Business and Finance: an Algebraic Approach, 3rd Edition
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Learning Unit Objectives

LU 4-1: Solving Equations for the Unknown.

1. Explain the basic procedures used to solve equations for the unknown.
2. List the five rules and the mechanical steps used to solve for the unknown in seven situations; know how to check the answers.

LU 4-2: Solving Word Problems for the Unknown.

1. List the steps for solving word problems.
2. Complete blueprint aids to solve word problems; check the solutions.

LU 4-3: Ratios and Proportions.

Understand ratios and proportions and solve for the unknown.

Basic Equation-Solving Procedure

Terminology

- **Expression** – A meaningful combination of numbers and letters called terms.
- **Equation** – A mathematical statement with an equal sign showing that a mathematical expression on the left equals the mathematical expression on the right.
- **Formula** – An equation that expresses in symbols a general fact, rule, or principle.
- **Variables** – *Equations use letters as symbols that represent one or more numbers. These symbols, usually a letter of the alphabet, are a **variable** that stand for a number.*
- **Constants** – *A fixed number such as a 3 or a -7.*

Variables and Constants Rules

1. If no number is in front of a letter, it is a 1:

$$B = 1B; C = 1C$$

2. If no sign is in front of a letter or number, it is a positive:

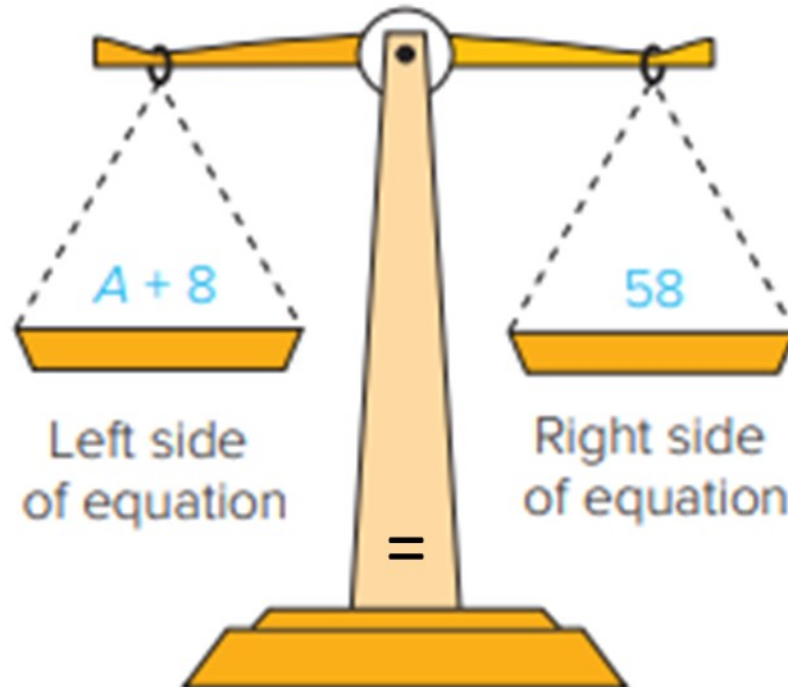
$$C = +C; 4 = +4$$

Solving Equations

- The meaning of the symbols $+$, $-$, $\times$ and $\div$ has not changed, however, some variations occur.
- You can write **A times B** as: $A \times B$, $A * B$ or AB .
- Key thing to remember is that to solve an equation, you must find a number that can replace the unknown in the equation and make it a true statement.

Solving Equations for the Unknown ¹

Equality in equations



Dick's age in 8 years will equal 58.

The variable "A" represents Dick's current age.

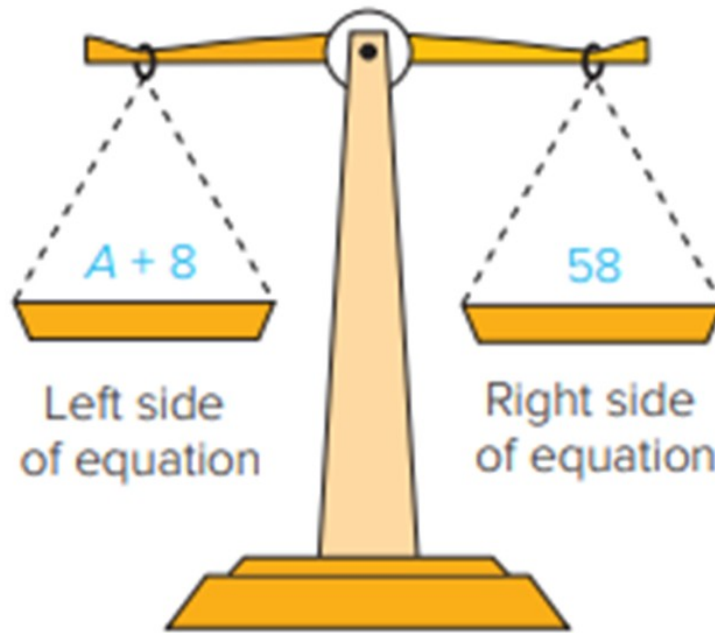
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Solving Equations for the Unknown ²

Verbal statement	Variable A (age)
Dick's age 8 years ago	$A - 8$
Dick's age 8 years from today	$A + 8$
Four times Dick's age	$4A$
One-fifth Dick's age	$A/5$

Solving for the Unknown Rule

- Whatever you do to one side of an equation, you must do to the other side.
- Perform operations of addition, subtraction, multiplication or division to both sides of the equation to keep the equation balanced.



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Opposite Process Rule ¹

If an equation indicates a process such as addition, subtraction, multiplication, or division, solve for the unknown or variable by using the opposite process.

For example:

- If the equation process is **addition**, solve for the unknown using **subtraction**.

If + then –

If – then +

If $\times$ then $\div$

If $\div$ then $\times$

Opposite Process Rule ²

EXAMPLE

$$A + 8 = 58$$

Dick's age A plus 8 equals 58.

Mechanical steps

$$\begin{array}{r} A + 8 = 58 \\ - 8 \quad - 8 \\ \hline A = 50 \end{array}$$

Explanation

8 is subtracted from *both* sides of equation to isolate variable A on the left.

Check

$$\begin{array}{l} 50 + 8 = 58 \\ 58 = 58 \end{array}$$

EXAMPLE

$$B - 50 = 80$$

Some number B less 50 equals 80.

Mechanical steps

$$\begin{array}{r} B - 50 = 80 \\ + 50 \quad + 50 \\ \hline B = 130 \end{array}$$

Explanation

50 is added to *both* sides to isolate variable B on the left.

Check

$$\begin{array}{l} 130 - 50 = 80 \\ 80 = 80 \end{array}$$

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Opposite Process Rule ³

Example

$$A + 8 = 58$$

Dick's age A plus 8 equals 58.

Mechanical steps

$$\begin{array}{r} A + 8 = 58 \\ - 8 \quad - 8 \\ \hline A = 50 \end{array}$$

Explanation

8 is subtracted from *both* sides of equation to isolate variable A on the left.

Check

$$\begin{array}{l} 50 + 8 = 58 \\ 58 = 58 \end{array}$$

Example

$$B - 50 = 80$$

Some number B less 50 equals 80.

Mechanical steps

$$\begin{array}{r} B - 50 = 80 \\ + 50 \quad + 50 \\ \hline B = 130 \end{array}$$

Explanation

50 is added to *both* sides to isolate variable B on the left.

Check

$$\begin{array}{l} 130 - 50 = 80 \\ 80 = 80 \end{array}$$

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Equation Equality Rule ¹

You can **add or subtract, multiply or divide** (except zero) the same quantity or number to both sides of the equation without affecting the equality of the equation.

For example:

- If the equation process is **multiplication**, solve for the unknown using **division**.
- Dividing both sides by 7 does not change the equality of the equation.

Equation Equality Rule ²

Example

$$7G = 35$$

Some number G times
7 equals 35.

Mechanical steps

$$7G = 35$$

$$\frac{7G}{7} = \frac{35}{7}$$

$$G = 5$$

Explanation

By dividing both sides by
7, G equals 5.

Check

$$7(5) = 35$$

$$35 = 35$$

Example

$$\frac{V}{5} = 70$$

Some number V
divided by 5
equals 70.

Mechanical steps

$$\frac{V}{5} = 70$$

$$8 \left(\frac{V}{5} \right) = 70(5)$$

$$V = 350$$

Explanation

By multiplying both sides by
5, V is equal to 350.

Check

$$\frac{350}{5} = 70$$

$$70 = 70$$

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Multiple Processes Rule ¹

When solving for an unknown that involves more than one process, do the **addition and subtraction before the multiplication and division.**

Multiple Processes Rule ²

Example

$$\frac{H}{4} + 2 = 5$$

When we divide unknown H by 4 and add the result to 2, the answer is 5.

Mechanical steps

$$\frac{H}{4} + 2 = 5$$

$$\frac{H}{4} + 2 = 5$$

$$\underline{-2} \quad \underline{-2}$$

$$\frac{H}{4} = 3$$

$$4\left(\frac{H}{4}\right) = 4(3)$$

$$H = 12$$

Explanation

1. Move constant to right side by subtracting 2 from both sides.
2. To isolate H , which is divided by 4, we do the opposite process and multiply 4 times *both* sides of the equation.

Check

$$\frac{12}{4} + 2 = 5$$

$$3 + 2 = 5$$

$$5 = 5$$

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Distributive Rule ¹

When equations contain parentheses (which indicates grouping together):

1. Solve for the unknown by multiplying each item inside the parentheses by the number or letter just outside the parentheses. **This is known as the distributive rule.**
2. Then continue to solve for the unknown with the opposite process used in the equation. Do the addition and subtraction first; then do the multiplication and division.

Distributive Rule ²

Example

$$5(P - 4) = 20$$

The unknown P less 4. multiplied by 5 equals 20.

Mechanical steps

$$5(P - 4) = 20$$

$$5P - 20 = 20$$

$$\begin{array}{rcl} & +20 & +20 \\ \hline 5P & & 40 \\ \hline 5 & = & 5 \end{array}$$

$$P = 8$$

Explanation

1. Parentheses tell us that everything inside parentheses is multiplied by 5. Multiply 5 by P and 5 by -4 .
2. Add 20 to both sides to isolate $5P$ on left.
3. To remove 5 in from of P , divide both sides by 5 to result in P equals 8.

Check

$$5(8 - 4) = 20$$

$$5(4) = 20$$

$$20 = 20$$

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Like Unknowns Rule

To solve equations with like unknowns:

1. Combine the unknowns.
2. Then solve with the opposite process used in the equation.

Example

$$4A + A = 20$$

Mechanical steps

$$4A + A = 20$$

$$\frac{\cancel{5}A}{\cancel{5}} = \frac{20}{5}$$

$$A = 4$$

Explanation

To solve this equation: $4A + 1A = 5A$. Thus $5A = 20$. To solve for A , divide both sides by 5 , leaving A equals 4 .

Check

$$4(4) + 4 = 20$$

$$20 = 20$$

Solving Word Problems for the Unknown ¹

5 Steps

1. Carefully read the entire problem. You may have to read it several times.
2. Ask yourself: “What is the problem looking for?”
3. When you are sure what the problem is asking, let a variable represent the unknown. If the problem has more than one unknown, represent the second unknown in terms of the same variable. For example, if the problem has two unknowns, Y is one unknown. Assuming the second unknown is 4 times the first unknown, it would be represented by $4Y$.
4. Visualize the relationship between the unknowns and variables. Then set up an equation to solve for the unknown(s).
5. Check your results to ensure accuracy.

Solving Word Problems for the Unknown ²

Blueprint aid

Helps organize unknown and given information and develop equation to solve word problems.

Unknown(s)	Variable(s)	Relationship

Solving Word Problems for the Unknowns ³

Example:

ICM Company sold 4 times as many computers as Ring Company. The difference in their sales is 27. How many computers of each company were sold?

Solution:

 Blueprint	Unknown(s)	Variable(s)	Relationship
	ICM Ring	$4C$ C	$4C$ $- C$ $\hline 27$

Mechanical steps

$$4C - C = 27$$

$$\frac{4C}{\cancel{4}} = \frac{27}{\cancel{3}}$$

$$C = 9$$

Ring = **9** computers

$$\text{ICM} = 4(9)$$

= **36** computers

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Solving Word Problems for the Unknowns 4

Example:

Andy sold watches for \$9 and alarm clocks for \$5 at a flea market. Total sales for 35 watches and alarm clocks were \$287. How many of each did Andy sell? What were the total dollar sales of each?

Solution:

	Unknown(s)	Variable(s)	Relationship
BLUEPRINT	ICM Ring	4C C	$ \begin{array}{r} 4C \\ - C \\ \hline 27 \end{array} $

Mechanical steps

$$\begin{array}{rcl}
 9W + 5(35 - W) & = & 287 \\
 9W + 175 - 5W & = & 287 \\
 4W + 175 & = & 287 \\
 \underline{- 175} & & \underline{- 175} \\
 4W & = & 112 \\
 \underline{A} & & \underline{4} \\
 W & = & 28 \\
 \text{Watches} & = & 28 \\
 \text{Clocks} & = & 35 - 28 = 7
 \end{array}$$

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Textbook Problem 4-2

Problem Statement:

Solve for the unknown from the following equation: *LU 4-1(2)*

$$A + 64 = 98$$

Solution:

$$\begin{array}{rcl} A + 64 & = & 98 \\ - 64 & & - 64 \\ \hline A & = & 34 \end{array}$$

Check:

$$A + 64 = 98$$

$$34 + 64 = 98$$

$$98 = 98$$

Textbook Problem 4-5

Problem Statement:

Solve for the unknown from the following equation: *LU 4-1(2)*

$$5Y = 75$$

Solution:

$$\begin{array}{rcl} 5Y & = & 75 \\ \hline \frac{\cancel{5}Y}{\cancel{5}} & = & \frac{75}{5} \\ \hline y & = & 15 \end{array}$$

Check:

$$\begin{aligned} 5Y &= 75 \\ 5(15) &= 75 \\ 75 &= 75 \end{aligned}$$

Textbook Problem 4-9

Problem Statement:

Solve for the unknown from the following equation: *LU 4-1(2)*

$$4(P - 9) = 64$$

Solution:

$$\begin{array}{rcl} 4P - 36 & = & 64 \\ \underline{+ 36} & & \underline{+ 36} \\ 4P & = & 100 \\ \underline{\cancel{4}} & & \underline{4} \\ P & = & 25 \end{array}$$

Check:

$$4(P - 9) = 64$$

$$4(25 - 9) = 64$$

$$4(16) = 64$$

$$64 = 64$$

Textbook Problem 4-15

Problem Statement:

Jessica and Josh are selling Entertainment Books to raise money for the art room at their school. One book sells for \$15. Jessica received the prize for selling the most books in the school. Jessica sold 15 times more books than Josh. Together they sold 256 books. How many did each one of them sell?: *LU 4-2(1)*

Solution:

Unknown(s)	Variable(s)	Relationship
Jessica Josh	15X X	$\begin{array}{r} 15X \\ + X \\ \hline 256 \text{ books} \end{array}$

$$15X + X = 256$$

$$16X = 256$$

$$X = 16(\text{Josh})$$

$$16 \times 15 = 240(\text{Jessica})$$

End of Main Content



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